Web Development Technical Assessment

Project: StockFlow - Inventory & Order Management Platform

Duration: 14 Days | Full-Stack Application

1. Project Brief & Core Roles

Project Overview

Build a full-stack Inventory & Order Management Platform that allows a business to manage products, inventory, suppliers, customers, and orders from a centralized platform. Staff maintain stock levels, process customer orders, and monitor inventory, while administrators manage the overall platform.

Primary User Roles

- **Administrator:** Manage and monitor the overall platform.
- **Staff:** Manage products, inventory, suppliers, and customer orders.
- **Customer:** Browse products and place/manage orders.

Development Flexibility

You are free to choose any programming language, framework, database, libraries, tools, or development environment you are comfortable with.

2. Assessment Rules & Duration

Development Tools & AI Policy

Permitted: AI assistants, AI coding tools, AI agents, documentation, Stack Overflow, tutorials, open-source libraries.

Rule: The final application must not contain an AI-powered feature. AI tools may be used as development assistance only. The submitted project should be a conventional web application.

Evaluation & Timeline

Explainability: You must be able to understand and explain the implementation submitted.

Duration & Timeline: Maximum Duration: 14 Days from the date shared. You may complete and submit earlier.

3. Detailed User Roles & Capabilities

Customer Capabilities

- Register and login
- Manage profile
- Browse products
- Search and filter products
- View product details
- Add products to cart & update quantity
- Place orders
- View order history & details
- Cancel eligible orders

Staff Capabilities

- Login
- Manage products, categories, suppliers
- Manage inventory & stock levels
- View customers
- Create/process orders
- Update order status
- View inventory history

Administrator Capabilities

Dashboard statistics, manage customers, manage staff, manage products, manage categories, manage suppliers, manage inventory, manage orders, activate/deactivate accounts, monitor platform records.

4. Customer Registration & Authentication

Registration Requirements

Minimum fields: Name, Email, Phone number, Password.

Input Validation

The application should handle required fields, invalid email, duplicate email, invalid password, and invalid phone number.

Role-Based Authentication

Securely distinguish between Customer, Staff, and Administrator. Users should only access functionality permitted for their role (e.g., customers cannot access inventory or admin features).

5. Product Catalog, Search & Filtering

Product Catalog

Customers and authorized staff view products. Examples: Laptop, Keyboard, Mouse, Monitor, Headphones, Smartphone, Printer, USB Cable.

Attributes: Product name, description, category, price, stock quantity, SKU/product code, supplier, image, status.

Search & Filters

Customers should be able to search and filter products by:

- Category & Price range
- Availability & Supplier
- Product status

6. Product Details & Supplier Management

Product Details Example

Wireless Keyboard
Price: ₹1,499 | Category: Accessories
Availability: In Stock | Stock: 24 units

The product page should provide an option to add the product to the cart.

Supplier Information

Authorized staff manage supplier info:

- Supplier name & Contact person
- Email, Phone, & Address
- Products supplied & Status

7. Inventory Management & Stock Tracking

Staff view current stock, add stock, reduce stock, update stock, view low-stock products, and manage availability.

Inventory Status Threshold Example

Product: Wireless Keyboard | **Current Stock:** 25 | **Low Stock Threshold:** 5 | **Status:** In Stock

Threshold reached: Current Stock: 4 → Status: Low Stock
Zero stock: Current Stock: 0 → Status: Out of Stock

8. Cart & Order Creation

Cart Actions & Totals Calculation

Browse products, view details, add to cart, change quantities, remove products, view total amount, confirm order.

Example Summary: Wireless Keyboard (₹1,499 × 2) + Wireless Mouse (₹799 × 1) = Subtotal ₹3,797, Discount ₹200, **Total ₹3,597**.

9. Critical Requirement - Inventory Consistency

Core Technical Challenge: Suppose Product Wireless Keyboard has Available Stock = 5. Customer A orders Quantity = 3 (remaining stock becomes 2). If Customer B attempts to purchase Quantity = 4, the system **must reject the order**.

The system must prevent negative stock scenarios (e.g., Buying 4 and 4 resulting in -3). The application must correctly handle concurrent order requests, backend logic, database operations, validation, and transaction handling.

10. Order Status & Staff Order Management

Workflow: PENDING → CONFIRMED → PROCESSING → READY → COMPLETED (or CANCELLED). Prevent invalid transitions.

Customer	Order	Date	Amount	Status
Rahul	#ORD001	15 Oct	₹3,597	Confirmed
Anu	#ORD002	15 Oct	₹1,999	Processing
Vishnu	#ORD003	16 Oct	₹899	Completed

11. Customer Orders, Cancellation & Stock Restoration

Customers view upcoming/current orders, order history, and status. Cancelling eligible orders restores stock appropriately (e.g., Initial 10 − 3 = 7 + cancelled 3 = 10), avoiding duplicate restorations on repeated attempts.

12. Inventory History & Stock Movement

Product	Action	Quantity	Date
Keyboard	Stock Added	+20	10 Oct
Keyboard	Order	-3	11 Oct
Keyboard	Order Cancelled	+3	12 Oct

13. Admin Dashboard & Management

Platform Statistics: Products (450), Customers (820), Suppliers (35), Orders (1,250), Pending (42), Completed (980), Cancelled (228), Low Stock (18), Out of Stock (7).

Controls: Manage customers, staff, products, categories, suppliers, inventory, orders, and activate/deactivate accounts.

14. Technical Architecture, Security & Database

Database & Backend: Relational/non-relational database with entities: Users, Customers, Staff, Products, Categories, Suppliers, Inventory, Transactions, Cart, CartItems, Orders, OrderItems.

Security: Hashed passwords, protected routes, role authorization, input validation, secure configuration, error handling.

15. Deliverables & 16. Assessment Weighting

GitHub, README, Deployment & Video: Complete source code, live URL, and 5-10 min screen recording demonstrating customer/staff/admin flows, stock deduction, and edge cases.

Category	Weight
Functional Requirements / UI/UX / Backend / Business Logic	65%
Database / Auth / Code Quality / Git & Docs	35%